

H1100 Hardened S46500 Stainless Steel

H1100 hardened S46500 stainless steel is S46500 stainless steel in the hardened (H) condition. It has the second lowest strength compared to the other variants of S46500 stainless steel.

The graph bars on the material properties cards below compare H1100 hardened S46500 stainless steel to: wrought precipitation-hardening stainless steels (top), all iron alloys (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Elastic (Young's, Tensile) Modulus

190 GPa

28 x 10⁶ psi

Elongation at Break

6.8 %

Fatigue Strength

560 MPa

81 x 10³ psi

Poisson's Ratio

0.28		
Rockwell C Hardness		
34		
Shear Modulus		
75 GPa		11 x 10 ⁶ psi
Shear Strength		
830 MPa		120 x 10 ³ psi
Tensile Strength: Ultimate (UTS)		
1410 MPa		210 x 10 ³ psi
Tensile Strength: Yield (Proof)		
1120 MPa		160 x 10 ³ psi

Thermal Properties

Latent Heat of Fusion		
280 J/g		
Maximum Temperature: Corrosion		
680 °C		1260 °F
Maximum Temperature: Mechanical		

780 °C	1430 °F
Melting Completion (Liquidus)	
1450 °C	2650 °F
Melting Onset (Solidus)	
1410 °C	2570 °F
Specific Heat Capacity	
470 J/kg-K	0.11 BTU/lb-°F
Thermal Expansion	
11 µm/m-K	

Otherwise Unclassified Properties

Base Metal Price	
15 % relative	
Density	
7.9 g/cm ³	490 lb/ft ³
Embodied Carbon	
3.6 kg CO ₂ /kg material	
Embodied Energy	

51 MJ/kg

22 x 10³ BTU/lb

Embodied Water

120 L/kg

14 gal/lb

Common Calculations

PREN (Pitting Resistance)

15

Resilience: Ultimate (Unit Rupture Work)

90 MJ/m³

Resilience: Unit (Modulus of Resilience)

3240 kJ/m³

Stiffness to Weight: Axial

14 points

Stiffness to Weight: Bending

25 points

Strength to Weight: Axial

50 points

Strength to Weight: Bending

36 points

Thermal Shock Resistance

49 points

Alloy Composition

Iron (Fe)	72.6 to 76.1
Chromium (Cr)	11 to 12.5
Nickel (Ni)	10.7 to 11.3
Titanium (Ti)	1.5 to 1.8
Molybdenum (Mo)	0.75 to 1.3
Manganese (Mn)	0 to 0.25
Silicon (Si)	0 to 0.25
Carbon (C)	0 to 0.020
Phosphorus (P)	0 to 0.015
Sulfur (S)	0 to 0.010
Nitrogen (N)	0 to 0.010

All values are % weight. Ranges represent what is permitted under applicable standards.

Followup Questions

How are the material properties defined?

How do I compare this exact variant to another material?

Further Reading

ASTM A693: Standard Specification for Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip

Welding Metallurgy of Stainless Steels, Erich Folkhard et al., 2012

ASTM A959: Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels

Properties and Selection: Irons, Steels and High Performance Alloys, ASM Handbook vol. 1, ASM International, 1993

Corrosion of Stainless Steels, A. John Sedriks, 1996

ASM Specialty Handbook: Stainless Steels, J. R. Davis (editor), 1994

Advances in Stainless Steels, Baldev Raj et al. (editors), 2010

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